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Superstore Dataset Analysis Procedure

1. Project Overview

This project was conducted to analyze Superstore's sales performance, regional performance, identify business opportunities and assess profitability. The objective was to transform raw sales data into actionable business insights through data cleaning, feature engineering, exploratory data analysis, and dashboard development in Power BI.

The analysis was conducted using a structured Business Intelligence workflow consisting of:

1. Business Understanding
2. Data Collection and Loading
3. Data Cleaning and Preparation
4. Exploratory Data Analysis (EDA)
5. Data Modeling
6. DAX Measure Development
7. Dashboard Design
8. Insight Generation
9. Business Recommendations

2. Data Sources

The project utilized the following datasets:

- Superstore.csv

Source: https://www.kaggle.com/datasets/vivek468/superstore-dataset-final

The dataset was imported into Power BI for analysis.

Steps: Get data > Text/CSV > Browse documents > Select Superstore CSV

3. Data Cleaning

The data was transformed in Power Query, and the following data-cleaning procedures were performed:

Data Validation

The dataset was reviewed for data quality issues before analysis. The following fields were validated:

Field	Data Type
Order Date	Date
Ship Date	Date
Sales	Currency, two decimal places
Profit	Currency, two decimal places
Quantity	Whole Number
Discount	Decimal Number/ Percentage

Formatting

- Date columns were formatted as dates (dd/mm/yyyy).
- Ensured all currency fields were formatted in Dollars and rounded to two decimal places.
- All text columns were formatted as text and the TRIM and CLEAN functions were applied.

Missing Values

- Columns were searched for blank and missing values. No missing values were identified.

Duplicate Records

- All columns were checked for duplicate records using the “Remove duplicates” and the “Keep Rows/Keep duplicates” button. No duplicates were identified in this dataset.

Customers ordered different items multiple times with the same order number. Customers also ordered various items on different days.

Relevant Columns

- The country column was removed since all the data was from only one country, the US.
- The postal code column was removed as it held no analysis value.
- The Days To Ship column was added to the table using the DATEDIFF() function.

Steps: Add New Column > Days to Ship = DATEDIFF(Order Date, Ship Date, Days)

4. Exploratory Data Analysis

Creating tables

- The Date table was created using the New Table button and the CALENDARAUTO() function.
- The Calculations table was created to house all calculations and DAX measures.

Creating DAX Measures

- DAX measures were created using the following formulas:

Measure	Formula/Function	Total
Total Sales	Total Sales = SUM(Superstore[Sales])	$2.30M
Total Profit	Total Profit = SUM(Superstore[Profit])	$286.40K
Total Orders	Total Orders = DISTINCTCOUNT(Superstore[Order ID])	5009
Average Sales	Average Sales = AVERAGE(Superstore[Sales])	$229.86
Profit Margin	Profit Margin = DIVIDE([Total Profit],[Total Sales],0)	12.47%
Overall **Average Shipping Days**	Average Shipping Days = AVERAGE(Superstore[Days To Ship])	4 days
Total Customers	Total Customers = DISTINCTCOUNT(Superstore[Customer ID])	793

Total Products	Total Products = DISTINCTCOUNT(Superstore[Product ID])	1862
Average Standard Shipping Days	Avg. Standard Shipping Days = CALCULATE(AVERAGE(Superstore[Days To Ship]), Superstore[Ship Mode] = "Standard Class")	5 days

Creating Discount Groups

The discount groups were required for the discount analysis. The groups were created using the following steps:

- Select the discount field under the Superstore table's dropdown menu in the Data pane
- Select "Column tools" > Click "Data groups" > Select "New data Groups"
- Name: Discount groups > Group type: List
- This shows a list of discount values in the "Ungrouped values"
- Group the values according to desired groups and label them:

Discount Group	Group Title
0	0%
0.1–0.2	10% to 20%
0.3–0.45	30% to 45%
0.5–0.8	+50%

- The "Discount Groups" field is created under the Superstore table's dropdown menu in the Data pane

5. Analysis Performed

The following analysis were performed, resulting in the below charts:

Sales, Discount, and Shipping Analysis

Metrics	Chart type
Total Sales vs. Average Sales	Line Chart
Total Profit vs. Profit Margin	Line Chart
Total Sales, Profit, and Profit Margin by Region	Line and Clustered Column Chart
Sales Density by State	Map
Top 10 States Revenue and Profit	Clustered Bar Chart
Top 10 Cities Revenue and Profit	Clustered Bar Chart
Revenue and Profit Margin by Customer Segment	Line and Clustered Column Chart
Total Sales, Profit, and Profit Margin by Product Category	Line and Clustered Column Chart
Top 10 Sub-Categories Revenue and Profit	Clustered Bar Chart
Revenue by Shipping Mode	Pie Chart
Revenue and Profit by Discount Groups	Line and Clustered Column Chart

The charts format and colors were edited using the “Format visual” settings in the Visualizations pane.

6. Dashboard Development

Three interactive dashboards containing KPI cards, slicers, charts, and tables were created:

a) Executive Dashboard

- KPI Cards included:
    - Total Sales
    - Total Profit
    - Profit Margin

    - Total Orders
    - Average Sales
- Slicers included:
    - Date
    - Region
    - Product Category
    - Customer Segment
    - State
- Charts included:
    - Total Sales vs. Average Sales
    - Total Profit vs. Profit Margin
    - Total Sales, Profit, and Profit Margin by Region
    - Sales Density by State

b) Detailed Dashboard

- KPI Cards included:
    - Total Customers
    - Total Products
    - Overall Average Shipping Days
    - Average Standard Shipping Days
- Slicers included:
    - Shipping Mode
    - City
- Charts included:
    - Top 10 States Revenue and Profit
    - Top 10 Cities Revenue and Profit
    - Revenue and Profit Margin by Customer Segment
    - Total Sales, Profit, and Profit Margin by Product Category
    - Top 10 Sub-Categories Revenue and Profit
    - Revenue by Shipping Mode

c) Sales Tables Dashboard

Charts included:

- Revenue and Profit by Discount Groups

Tables included:

- Total Sales, Profit, and Profit Margin by Per Year
- Sales by Customer Name
- Sales by Product Name

7. Tools Used

- Power BI
- Power Query – Extract, Transform, and Load (ETL)
- Visuals – Bar, column, pie, combo charts, maps and matrix tables
- Slicers
- New Measures
- New Table
- New Column
- DISTINCTCOUNT()
- SUM() and AVERAGE()
- DIVIDE()
- CALCULATE(AVERAGE())
- Grouping feature – Column tools > Data groups

8. Project Outcome

The project successfully transformed transactional sales data into actionable business insights. Key achievements include:

- Identifying growth trends in sales and profit.

- Highlighting the company's most profitable regions, products, and customer segments.
- Identifying underperforming markets.
- Evaluating shipping methods.
- Assessing the relationship between discounts and profitability.
- Developing data-driven recommendations to improve business performance.
- The resulting dashboard provides management with an effective tool for monitoring performance, identifying risks and opportunities, and supporting strategic decision-making.